APPROXIMATE ESTIMATE

Lecture III

APPROXIMATE ESTIMATE

- Also called preliminary or rough estimate
- Detailed drawings are not required
- A line sketch may be required
- Approximate estimate for main parts of a project as for buildings services like sanitary , water supply, drainage, electrification, boundary walls, roads, cost of land are done separately.

PURPOSE OF APPROXIMATE ESTIMATE

- To investigate the financial feasibility of the project.
- To save time and money – why prepare detailed estimate if the project is not approved.
- Through the preliminary study, if the investment return outweighs the cost, one can proceed.
- Adjustment of planning – to avoid estimating costs for all alternatives of the design.
- To obtain administrative approval – approximate estimate + report + site plan/layout plan is submitted.
- For insurance & tax schedule purpose, the value of a property is drawn up from approximate estimate.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

- Plinth Area Basis/ Area method
- Cubic rate or cubic meter method
- Approximate quantities with Bill method
- Service unit/unit rate method
- Cost comparison method
- Cost from materials and labour
- Approximate cost of water supply, sanitation, electrification works.
- Based on CBRI formula

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

1

PLINTH AREA BASIS

Need to calculate the plinth area first

Plinth area may also have to be worked our from floor area/carpet area/covered area/rentable area of a building.

Plinth area includes	Plinth area doesn't include
Stair cover and lift well including landing, machine room	Area of loft, balcony
Internal shaft for sanitary installations & garbage chute (not exceeding 2 sq m in area)	Area of architectural feature like band or cornice. Area of vertical sun breaker or box projecting out. Towers, turrets, domes projecting above terrace
Area of the floor level excluding plinth offsets	Cantilever porch, open platform, terrace at floor one
If building columns project beyond cladding, then external face of cladding	Spiral staircase without landing
Area of porch other than cantilevered	Additional floor for seating in assembly or theatres, auditoriums.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

a

FLOOR AREA BASIS

Floor Area = Plinth Area – Area of the walls.

In the calculation of wall area, thickness of the wall is taken inclusive of finishing & dado if the height of the finish is more than 1 m from the floor finish.

The following is to be included in the wall area :-

- Door and other openings in the wall
- Intermediate pillars, supports or any other obstruction within the plinth area irrespective of their location.
- Pilaster along wall exceeding 300 sq cm in area.
- Flues within the wall
- Fire place projecting beyond the face of the wall
- Built in cupboards, almirahs, shelves within a height of 2.2 m from floor.

- Mezzanine , garage are measured separately.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

b

CARPET AREA BASIS

Carpet Area = Floor Area – the following:

- Verandah
- Corridor/ passage
- Entrance hall/porch
- Staircase and stair cover
- Bathroom, unusable area of living
- Kitchen/pantry
- Store
- Canteen
- Shaft and machine room for lift
- AC duct & plant room
- Shaft for sanitary piping.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

C

COVERED AREA

It is the ground area covered by the building immediately above the plinth level.

The area covered by the following in the open area is excluded –

- Watchman's booth, pump house, garbage shaft, electric cabin.
- Uncovered staircase, ramps, areas covered by chajja, slide, swing, portico
- Garden, plant nursery, swimming pool (uncovered)
- Drainage, culvert, conduit, gutter and other services.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

d

RENTABLE AREA

For residential buildings, the area shall be the carpet area along with –

I. The carpet area of kitchen, pantry, store, lavatory, bathroom

II. 50% of carpet area of unglazed and 100% of glazed verandah

Will not include –

- Storage space on top
- Landings of staircase
- Under the first landing and waist slab on floor one
- 50% of the area of balconies.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

e

CIRCULATION AREA OR FREE SPACE AREA

It includes – verandahs, balconies, passages, corridors, porches, entrance halls, staircases, mumties, shafts for lifts.

It may be :

- Horizontal circulation area – 10% to 15% of the building area is sufficient.
- Vertical circulation area – 4% to 5% of the area (lifts, staircases)

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

ESTIMATED COST OF A PROPOSED BUILDING ON PLINTH AREA BASIS –

Estimated Cost = Plinth Area of the building X Plinth Area Rate from the locality for similar buildings constructed recently.

Plinth Area = Horizontal circulation area + Carpet area

OR

Floor area + Area of walls

✔ Square buildings are most economical as they give more floor area with less wall running length.

THUMB RULE FOR CALCULATING AREA OF WALLS

- 8% of floor area ⇒ R.C.C framed structure
- 15% to 17% of floor area ⇒ Ordinary buildings
- 25% of floor area ⇒ Thick walls for rectangular shaped buildings

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

g

PLINTH AREA RATES DEPENDS UPON –

- General specifications – quality of materials used
- Price level – m^2 rate of buildings in locality
- Shape of building
- Design of a building – structure features, thickness of walls etc.
- Arrangement of rooms, circulation areas and provisions made for sanitary blocks
    - Attached toilets rate up price
    - Compact planning (mere walls) cost more than spacious planning
- Location

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

2

CUBIC RATE OR **CUBIC METER METHOD** –

The use of cubic meter volume in estimating is more accurate than using just plinth area as cost depends on height also

Estimated Cost = Volume or cubic content of the building X rate per cubic volume of similar buildings in the locality constructed recently

$$\text{Cubic Volume} = \underbrace{\text{Length X Breadth}}_{\text{Plinth Area}} \text{ X } \underbrace{\text{Height}}_{\text{It may be considered from}}$$

- Ground level
- Top of foundation
- Half of the depth of foundation
- 15 cms below the finished surface of lowest floor

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

CUBIC RATE OR **CUBIC METER METHOD –**

Cost of building depends on depth of foundation.

As per meter cost below ground level is not same as per meter cost above ground level, it is best to calculate height from half depth of foundation to top of roof (if not parapet) or to half height of parapet for flat roof, for sloped roof (height is taken from point midway between the ridge and the intersection of the roof with the wall)

For multi-storeyed – floor of one storey to another

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

3

APPROXIMATE QUANTITIES WITH BILL METHOD –

Multiply the total length of the walls of a building in running metre by the cost of construction per metre length of such a wall –

The total length of the wall is calculated from the plan, each section of wall separately. The cost of unit running metre of wall will include, excavation, foundation, concrete, brickwork, plastering, finishing, DPC, parapets etc; so its best to calculate it in parts.

- Cost of wall including foundation & surface finishing.
- Cost of woodwork.
- Cost of flooring.
- Cost of roofing.

All sub heads are added to get an approximate estimate. It is much more realistic, almost like detailed estimate.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

4

SERVICE UNIT / UNIT RATE METHOD –

All costs of a unit quantity are calculated like per classroom for a school building, per bed for a hospital are estimated and multiplied the cost per corresponding unit with number of units.

Can be derived from other similar government projects.

5

BAY METHOD –

<u>Estimated Cost</u> = Number of bays in proposed structure X Cost of one such bay

Bays are compartments or similar portions of a structure. Like if a structure is made of similar cabins or parts as in godowns, railway platform, factory shades etc.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

6

COST COMPARISON METHOD –

Approximate estimates are prepared for prototype structure or units consisting of several works after comparing with the past expenses for such works.

7

COST FROM MATERIALS AND LABOUR–

Approximate quantities of materials and labour required per sq. mt. for the plinth area of a proposed building are worked out with the help of empirical equations developed by CBRI Roorkee and multiplied by the plinth area of the building.

For example,

Bricks (% nos.) $\longrightarrow$ 2.26A + 66.8 …………………for single storeyed buildings

A = plinth area.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

8 APPROXIMATE COST FOR WATER SUPPLY, SANITATION AND ELECTRIFICATION WORKS–

Water supply = 4 to 5% of the estimated cost for building works.

Sanitary = 4 to 5% of the estimated cost for building works.

Electrification = 9% of the estimated cost for building works.

Electric fans, lights = 3% of the estimated cost for building works.

a COST OF WATER SUPPLY–

(4 to 5% of building structure)+400/- if simple street pipe connection (5mts from property line to OHT).

If OHT water is pumped from underground storage tank, add 1200/-
If pumped from a tubewell, add 1800/-
} added to 4 to 5% of building cost

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

b

COST OF SANITATION–

4 to 5% of building cost. In absence of sewer line, septic tank with a soak pit should be provided. Then 7 to 9% of building cost is needed.

c

COST OF ELECTRICAL WORKS–

9% of building cost. If cost of electric fans and lights is added, then it comes to 12%.

9

COST BREAKUP BASED ON CBRI FORMULA–

Cost on account of materials – 75 % of the total cost for materials and labour.

Cost on account of labour – 25% of the total cost for materials and labour.

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

Cost of bricks –18% of the cost of materials and labour.

Cost of cement – 16% of the cost of materials and labour.

Cost of steel – 11% of the cost of materials and labour.

Cost of sand – 4% of the cost of materials and labour.

Cost of aggregates – 9% of the cost of materials and labour.

Cost of lime – 4% of the cost of materials and labour.

Cost of surkhi – 0.4% of the cost of materials and labour.

Cost of timber – 12% of the cost of materials and labour.

Cost of paint/ primer – 0.2% of the cost of materials and labour.

Cost of door/ window fittings – 0.4% of the cost of materials and labour.

Total = 75%
Labour = 25%

DIFFERENT METHODS OF APPROXIMATE ESTIMATE

MATERIALS REQUIRED FOR A SINGLE STOREY BUILDING USING CBRI FORMULA –

Name of materials	Quantity of materials per sq. mt. of plinth area			Avg. recommendation per sq. mt.
	60sqm.	100sqm.	150sqm.	
Brick	@337 nos.	@293 nos.	@270.5 nos.	300 nos.
Cement	@3.5 bags	@3.11 bags	@3.14 bags	3 and half bags
Steel	@16 kgs	@18.16 kgs	@19.21 kgs	18 kgs
Sand	@3.5 cubic mt.	@0.4 cubic mt.	@0.42 cubic mt.	0.4 cubic mt.

COST BREAKUP-

Total cost 100% (excluding water supply, sanitation, electrification)

- Cost of foundation and plinth 18%
  - Cost up to GL 13%
  - Cost from GL to PL and up to DPC 5%
- Cost of brickwork from plinth to roof 23%
- Cost of RCC except roofing 6%
- Cost f roofing 20%
  - RCC roof 15%
  - Lime terracing 5%
- Flooring 6%
- Doors and windows 20%
- Plastering and finishing 6%

NUMERICALS

QUESTION 1-

A person constructs a building of plinth area equal to 100 sqm on a plot of land in certain locality at a cost of Rs. 95,000. The height of the building from ground level to the top of roof is 3.5 m and a parapet wall of height equal to 80 cm and is constructed on the terrace.

Determine the cost of a similar building of plinth area equal to 135 sqm to be constructed in the same locality by plinth area rate and also by 'volume rate'.

NUMERICALS

QUESTION 1-

SOLUTION-

By plinth area method:

For 100 sqm plinth area, cost = Rs. 95,000

Plinth area rate = Rs. 950 /- sq m

For 135 sqm, cost of similar building = Rs. 135 X 950 = Rs. 1,28,250

NUMERICALS

By volume method:

Considering height of the building from GL to half of parapet, cubic volume of existing building = 100sqm X (3.5+0.8/2m) = 390 cubic mt.

Cubic volume rate = Rs. 95000/390 = Rs. 243.59

Volume of similar building = 135sqm X (3.5+0.8/2m) = 526.5 cubic mt.

Cost of proposed building = Rs. 526.5 X 243.59 = Rs. 1,28,250.13 (same cost as derived in plinth area)

NUMERICALS

QUESTION 2-

Preliminary estimate required for school building for 500 students. Carpet area required per student is 1.20 sqm, area of corridor, verandah and washrooms is 20% of plinth area, walls 15% of plinth area.

Plinth area rate Rs. 1100 /sq m

Water supply = 5% of building cost.

Sanitation = 6% of building cost.

Electrification = 10% of building cost.

Approach road + boundary wall = 3% of building cost.

Contingency = 5%, Work charged/ establishment = 2.5%.

NUMERICALS

Carpet area for students = 1.2 X 500 = 600sqm.

P = Plinth area, P = carpet area + walls + corridor. Therefore, P = 600 + (P X 20/100) + (P X 15/00)

Multiplying by 100 on both sides, 100P = 60000 +20P+15P, equating this we get P = 923.08 sqm (plinth area)

Cost of building having 923.08 sqm plinth area @1100 = Rs. 10,15,388

Water supply @5% = Rs. 10,15,388 X (5/100) = Rs. 50,769.40

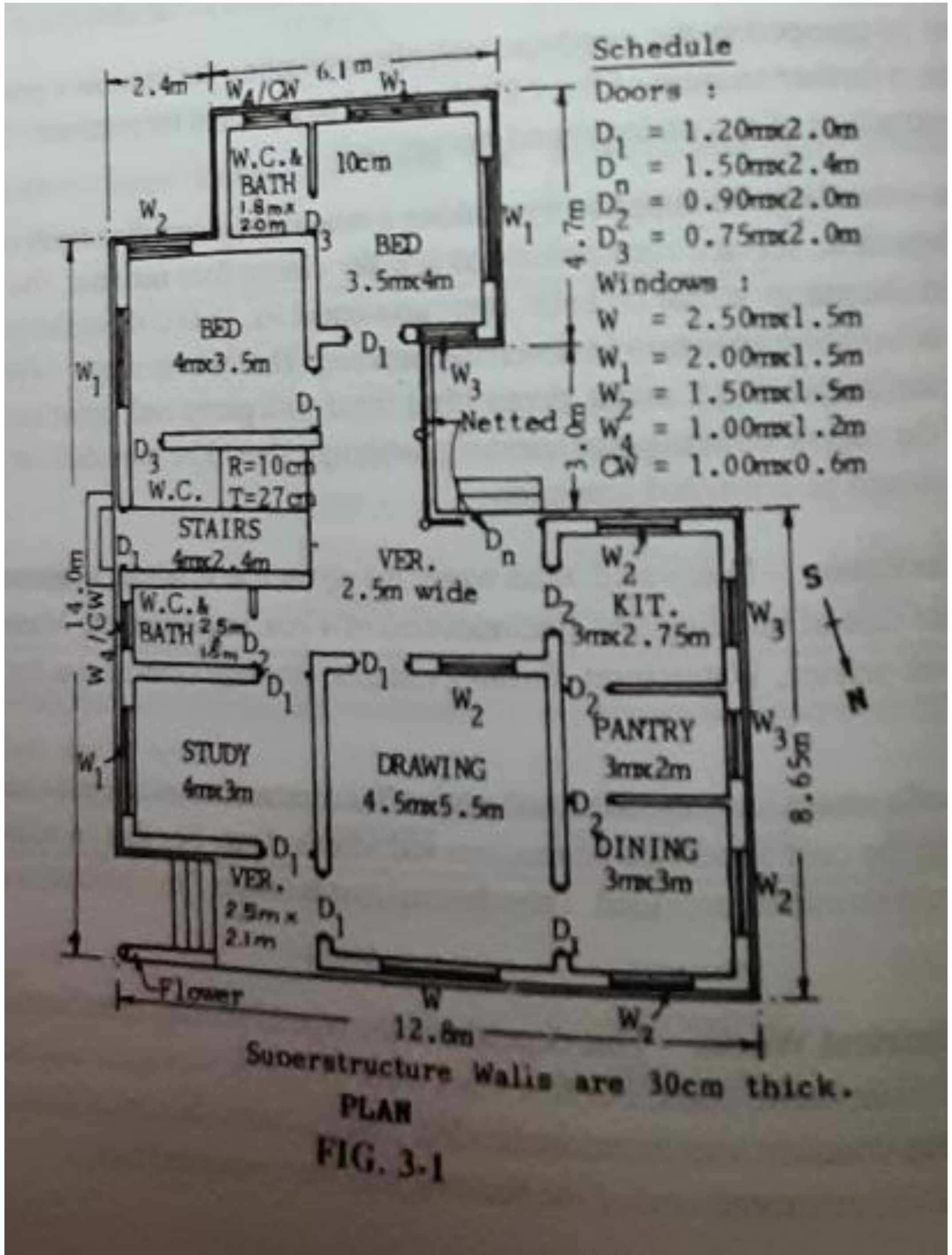
Sanitation @6% = Rs. 10,15,388 X (6/100) = Rs. 60,923. 28

Electrification @10% = Rs. 10,15,388 X (10/100) = Rs. 1,01,538.80

Approach road, boundary @3% = Rs. 10,15,388 X (3/100) = Rs. 30,461.64

Total = Rs. 12,59,081.12

Adding contingency 5% and establishment 2.5% to this, total cost comes to Rs. 13,53,512. 20



Estimate using Plinth area method @Rs. 2100/sqm

Calculating plinth area excluding plinth offset (offset = 0.05 m):

Front portion upto kitchen = 12.7x8.55=108.59 sq m

Middle portion from kitchen to verandah = 3.1x(2.5+0.3+4+0.3) = 22.01 sq m

Back portion = 4.6x(6+2.4) – (2.4x2.3) = 33.12 sqm

Total = 163.72 sq m

@Rs 2100/sq m cost of building = 163.72x2100 = Rs. 3,43,812